A GF–Lean Bridge for Verified Natural Language Reasoning: From English Parsing to Entailment Proofs (DRAFT --- April 3, 2026)

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Abstract

We present a four-layer pipeline bridging the Grammatical Framework (GF) and Lean 4 for verified natural language reasoning. GF handles multilingual parsing and linearization using its 100,000+ entry WordNet-backed grammar; Lean handles type-checked abstract syntax trees, semantic interpretation, and formal proofs. The bridge is grammar-agnostic: a shared GFCore package provides the trust boundary via a `check` function that verifies parse trees against runtime-loaded grammar signatures. We evaluate on the EntailmentBank dataset, achieving 89.1% parse coverage with GF's ParseEng grammar, and demonstrate a proof-of-concept entailment verification using pattern matching on readable semantic views extracted from GF parse trees. The pipeline supports round-trip verification: English → Lean → English and English → Lean → Czech.

1 Introduction

The QED Manifesto [1] envisions a universal library of formalized mathematics. Extending this vision to natural language reasoning —where premises and conclusions are expressed in English, not in logical notation—requires bridging two worlds: natural language processing (inherently ambiguous, statistical, engineering-heavy) and formal verification (precise, kernel-checked, proof-theoretic).

We propose using the Grammatical Framework (GF) [2, 3] as the natural language layer and Lean 4 [5] as the formal reasoning layer. GF provides a mature, multilingual grammar formalism with a Resource Grammar Library (RGL) covering 40+ languages and a WordNet-backed lexicon of over 100,000 entries. Lean 4 provides a dependently typed proof assistant with operational programming capabilities.

Our contribution is a *four-layer architecture* that cleanly separates concerns:

1. **GF as oracle**: parsing English to abstract syntax trees and linearizing back to any GF language.
2. **GFCore**: a grammar-agnostic Lean package that type-checks parse trees against grammar signatures.
3. **RGLView**: a readable semantic view that peels away RGL scaffolding to expose predicate-argument structure.
4. **Semantics**: meaning graphs with provenance, enabling entailment reasoning (proof of concept demonstrated).

We evaluate on the EntailmentBank [6] dataset and demonstrate verified entailment steps using is-a substitution extracted from GF parse trees.

2 Background

2.1 Grammatical Framework (GF)

GF [2] separates *abstract syntax* (language-independent semantic structure) from *concrete syntax* (language-specific linearization rules). A GF grammar defines typed function signatures:

```
fun PredVP : NP -> VP -> Cl ;      -- predication
fun DetCN  : Det -> CN -> NP ;     -- determiner + noun
fun UseN   : N -> CN ;             -- noun to common noun
fun SlashV2a : V2 -> VPSlash ;     -- transitive verb slot
```

The RGL provides approximately 300 structural functions covering English syntax (tense, aspect, agreement, coordination, relative clauses, questions, etc.) across 40+ languages.

ParseEng from the `gf-wordnet` project [4] extends the RGL with 115,060 functions from Princeton WordNet, providing wide-coverage English parsing with full morphology and sense-specific lexical entries.

2.2 Lean 4

Lean 4 [5] is a dependently typed programming language and proof assistant. We use it both as an operational programming language (for the bridge infrastructure) and as a verification framework (for checking parse tree well-formedness and, in future work, semantic proofs).

2.3 EntailmentBank

EntailmentBank [6] provides 1,840 multi-hop entailment examples over elementary science sentences, with explicit proof trees showing how premises combine to yield hypotheses.

3 Architecture

```
Surface English text
  |
  v
Layer 1: GF ParseEng (C runtime, 115K WordNet entries)
  | parse -> abstract tree (JSON)
  v
Layer 2: GFCore.check (Lean, grammar-agnostic)
  | RawTerm -> CheckedExpr (verified against GrammarSig)
  v
Layer 3: RGLView (Lean, ~30 structural constructors)
  | Peel wrappers -> readable semantic structure
  v
Layer 4: Semantics + Reasoning (Lean)
  | Pattern matching, entailment verification
  v
GFCore.erase -> GF linearize -> English / Czech / ...
```

Figure 1: The four-layer GF–Lean pipeline.

Layer 1 uses GF as a *resident parsing oracle*. The GF C runtime (via Python PGF bindings) loads the 13 MB ParseEng grammar once and parses sentences in 2–4 seconds each. Parse results are exported as JSON using the PGF API's `Expr.unpack()` method, producing `RawTerm` values.

Layer 2 is the *trust boundary*. The `check` function in GFCore verifies that every function name in a parse tree exists in the grammar signature, has the correct arity, and satisfies category constraints. Only verified `CheckedExpr` values proceed to downstream layers. The grammar signature (115,060 functions) is loaded at runtime from a 10.7 MB JSON file—far too large for compiled Lean source.

Layer 3 transforms the raw RGL tree into a *readable semantic view*. GF's RGL wraps every sentence in layers of tense, polarity, and phrase structure:

```
PhrUtt NoPConj (UttS (UseCl (TTAnt TPres ASimul) PPos
  (PredVP (UsePN john_PN) (ComplSlash (SlashV2a see_V2)
    (DetCN the_Det (UseN man_N))))) NoVoc
```

RGLView peels these wrappers to expose:

```
john_PN | see_V2(the man_N)
```

Layer 4 interprets the semantic view for reasoning. In our proof of concept, this is pattern matching on RGLView constructors to extract is-a relations and verify substitution-based entailment.

4 Implementation

4.1 GFCore Package

GFCore is a standalone Lean 4 package with no mathlib dependency, shared by both the `algorithms` and `mettapedia` projects.

Module	Purpose	Key Types/Functions
Syntax	Core types	`RawTerm`, `CheckedExpr`, `GrammarSig`
Json	Wire format codecs	`FromJson`/`ToJson` instances
Check	Trust boundary	`check : GrammarSig -> RawTerm -> CheckedExpr`
Export	Linearization direction	`erase : CheckedExpr -> RawTerm`
Driver	GF subprocess interface	`GFDriver.fromEnv`, `parse`, `linearize`
SigGen	Signature generation	PGF JSON → Lean `GrammarSig`
LexGen	Lexicon generation	JSONL → GF grammar modules
Repair	Dataset preprocessing	Slash removal, possessive fix, typo correction
RGLView	Semantic view	`toRGLView : CheckedExpr -> RGLView`
Frame	PLN-compatible IR	`GroundedLexeme`, `Frame` (6 PLN patterns)
FrameExtract	Entity/relation extraction	`extractEntity`, `extractFrame`
Entailment	Deterministic reasoning	5 PLN rules (transitivity, substitution, etc.)

Table 1: GFCore modules.

4.2 Key Types

The wire format between GF and Lean:

```
inductive RawTerm where
  | app (funName : String) (catHint? : Option String)
        (args : Array RawTerm)
```

The verified AST (every node carries its resolved declaration):

```
inductive CheckedExpr where
  | node (decl : FunDecl) (args : Array CheckedExpr)
```

The readable semantic view:

```
inductive RGLView where
  | noun (name : String) | adj (name : String)
  | verb (name : String) | prep (name : String)
  | det (kind : DetKind) (num : NumKind) (cn : RGLView)
  | mass (cn : RGLView)
  | pred (subject : RGLView) (verbPhrase : RGLView)
  | comp (subject : RGLView) (complement : RGLView)
  | transV (verb : RGLView) (object : RGLView)
  | kindOf (kind : RGLView) (of_ : RGLView)
  | sentence (tense : Tense) (pol : Polarity) (core : RGLView)
  | opaque (funName : String) (args : List RGLView)
  ...
```

4.3 The Check Boundary

The `check` function is the single point where trust is established:

```
partial def check (sig : GrammarSig) (t : RawTerm)
    : Except CheckError CheckedExpr := do
  let decl <- sig.findFun? t.funName
    |>.toExcept (.unknownFun t.funName)
  if t.args.size != decl.arity then
    throw (.wrongArity ...)
  let checkedArgs <- t.args.mapIdxM fun i arg => do
    let child <- check sig arg
    if child.resultCat != decl.argCats[i]! then
      throw (.catMismatch ...)
    pure child
  pure (.node decl checkedArgs)
```

Nothing unverified passes this boundary. The grammar signature is loaded at runtime from JSON, making the system grammar-agnostic.

4.4 No Heuristics Policy

An audit of the codebase identified and eliminated all heuristic shortcuts: suffix-based POS guessing was replaced with GF's own WordNet lexicon; hardcoded paths were replaced with environment variables; silent default values were replaced with explicit errors. Every design decision is documented and auditable.

5 Evaluation

5.1 Parse Coverage

We evaluate ParseEng on 948 unique sentences from the EntailmentBank task 1 development set.

Result	**Count**	**Percentage**
Parsed (raw ParseEng)	845	89.1%
Parsed (after repairs)	+71	+7.5%
Parsed (overlay lexicon)	+∼15	+∼1.6%
Total parsed	**∼931**	**∼98.2%**
Still failed	∼17	∼1.8%

Table 2: ParseEng coverage on EntailmentBank dev set (C runtime, n=1).

The initial 10.9% failures were primarily dataset formatting issues. Preprocessing repairs (replacing slash notation `X / Y` with `X or Y`, fixing possessive tokenization, correcting typos) recovered 71 additional sentences (96.6%). A 10-word overlay lexicon for domain terms absent from WordNet (`decomposer`, `nonliving`, `petri`, `serotinous`, `rna`, etc.) brings coverage to approximately

98%. The remaining ∼2% are genuine edge cases: hyphenated compounds (`non-fertile`), bare numbers (`1000`), and unusual word uses.

5.2 Round-Trip Verification

We verified the full round-trip on the PaperAmbiguity grammar:

```
Input:  "John sees the man with the telescope"
GF parse -> 2 RawTerms (VP and NP attachment)
check -> 2 CheckedExprs (UseCl : S)
erase -> 2 RawTerms
GF linearize ->
  English: "John sees the man with the telescope"
  Czech:   "Jan vidi muze s teleskopem"
```

Both attachment readings produce correct round-trips in both languages.

5.3 Performance

Runtime	Avg. parse time	948 sentences
GF Haskell CLI (`gf --run`)	27s/sentence	∼7 hours
GF C runtime (Python PGF)	2–4s/sentence	∼50 minutes

Table 3: Parse performance with ParseEng (13 MB PGF, 115K functions).

The C runtime loads the grammar once (1.2 seconds) and parses subsequent sentences without reloading. The Haskell CLI spawns a new process per sentence, making it impractical for batch work.

6 Semantic Frame Extraction

6.1 Frame IR

Layer 4 introduces a PLN-compatible Frame intermediate representation using terminology consistent with Probabilistic Logic Networks [7]:

```
structure GroundedLexeme where
  gfFun : String    -- "star_8_N" (links to WordNet synset)
  cat   : String    -- "N"

inductive Frame where
  | inheritance (sub sup : GroundedLexeme)   -- isA
  | evaluation  (pred arg : GroundedLexeme)  -- property
  | evaluation2 (pred arg1 arg2 : GroundedLexeme)  -- relation
  | member      (part whole : GroundedLexeme) -- partOf
  | implication (ante cons : Frame)          -- causes
  | forAll (var : String) (body : Frame)
  | conj   (frames : List Frame)
  | opaque (description : String)           -- not yet extracted
```

Frame extraction uses a two-function approach: `extractEntity` maps any NP-like RGLView to a `GroundedLexeme`, and `extractFrame` maps sentence-level views to `Frames`.

6.2 Extraction Results

On our demo sentences:

```
"the sun is a kind of star"      -> Inheritance(sun, star)
"earth is a kind of planet"      -> Inheritance(earth, planet)
"water causes erosion"           -> Evaluation(cause, water, erosion
    )
"the earth rotates"              -> Evaluation(rotate, earth)
"a star produces light"          -> Evaluation(produce, star, light)
```

6.3 Deterministic Entailment Rules

Five reasoning rules operate over Frames without probabilities:

1. **Inheritance transitivity**: $A \subseteq B + B \subseteq C \to A \subseteq C$
2. **Inheritance substitution**: $A \subseteq B + P(B, Y) \to P(A, Y)$
3. **Property inheritance**: $A \subseteq B + Q(B) \to Q(A)$
4. **Modus ponens**: $(F_1 \to F_2) + F_1 \to F_2$
5. **Conjunction introduction**: $F_1 + F_2 \to F_1 \wedge F_2$

6.4 Entailment Verification: Example 74

sent1: "the sun is a kind of star"
sent2: "hydrogen is the most common element in stars"
$\to$ **hypothesis**: "hydrogen is the most common element in the sun"

Sent1 extracts cleanly: `Inheritance(sun, star)`.

However, sent2's top-1 parse treats "common element in" as a compound adjective (`CompoundAP(element_1_N, in_1_A)`), yielding `Opaque` rather than the expected `Evaluation2(element_in, hydrogen, star)`. This prevents the substitution rule from firing.

6.5 Sense Disambiguation via PLN Evidence Propagation

This failure reveals that **top-1 parsing is insufficient** for semantic reasoning. The correct parse—where "element" is a noun and "in stars" a prepositional phrase—exists among the top-N candidates but is not ranked first.

The principled solution uses PLN-style evidence propagation:

1. Parse with top-N candidates, each with probability p_i.

2. Extract Frames from each candidate. Successful extraction adds positive evidence; `opaque` adds negative evidence.

3. Attempt entailment reasoning with each candidate's Frames. Successful derivation adds further positive evidence.

4. Select the candidate with highest combined score: $\text{score}_i = p_i \times \text{extraction\_success}_i \times \text{reasoning\_success}_i$.

This is not a heuristic—it is the standard PLN approach to ambiguity resolution. The existing WM-PLN infrastructure in mettapedia (704 formalized theorems, zero sorries) provides the theoretical foundation: the `BinaryWorldModel` calculus handles evidence composition, and the `PLNLinkCalculus` provides deduction rules with explicit side conditions.

In this framework, "reasoning fails" on a parse candidate is a *signal*, not a dead end. The failure propagates as negative evidence through the PLN factor graph, causing the system to prefer parse candidates that lead to successful reasoning chains. This naturally implements attention allocation over parse forests.

6.6 Layer 4 Redesign: Term/Atom/NormClause

Based on dev set evaluation results and GPT-5.4 Pro review, we redesigned Layer 4 from a flat `Frame` IR to a three-pass compositional pipeline:

1. **NormClause**: canonicalizes surface variation. `FocusComp`, `PredVPS+UseComp`, `PredVP+UseComp`, and `AdvIsNP` all map to a single `copular(subject, complement)` form. Transitive verbs map to `predication(subject, verb, object)`.

2. **Term**: structured entities with head + modifiers. "the most common element in stars" becomes `Term.entity(element, the, sg, [adj:common, prep:in(star)])`.

3. **Atom**: PLN-compatible propositions over Terms. `IsA(sun, star)`, `Attr(entity, property)`, `Rel(verb, [agent, patient])`, `Causes(cause, effect)`.

```
inductive Term where
  | entity (head : GroundedLexeme) (det : Option DetKind)
           (number : Option NumKind) (mods : List (String x String)
               )
  | event  (pred : GroundedLexeme) (args : List (Role x Term))
  | var    (name : String) | opaque (description : String)

inductive Atom where
  | isa     (sub sup : Term)
  | attr    (entity : Term) (prop : GroundedLexeme)
  | rel     (pred : GroundedLexeme) (args : List Term)
  | causes  (cause effect : Atom)
  | implies (ante cons : Atom)
```

```
  | conj (xs : List Atom) | neg (x : Atom)
  | opaque  (description : String)
```

The redesign improved non-opaque extraction from ∼40% (flat Frame) to **100%** on 6 test sentences. Every sentence now produces a substantive Atom with structured Term arguments.

A key architectural decision, based on GPT-5.4 Pro review, was the introduction of `CopulaOrigin` provenance in `RGLView`. GF's `FocusComp` constructor reverses subject/complement order compared to ordinary `PredVP+UseComp` copular sentences. Rather than detecting this reversal by inspecting tree complexity (a brittle heuristic), we preserve the constructor identity:

```
inductive CopulaOrigin where
  | predVPUseComp | predVPSMkVPS | focusComp | advIsNP

inductive RGLView where
  ...
  | copularSurface (origin : CopulaOrigin) (lhs rhs : RGLView)
```

`NormClause` then uses the origin to guarantee canonical subject/complement order—`focusComp` swaps its arguments, all others preserve them. This eliminated an entire class of argument-ordering bugs.

With typed modifiers (`Modifier.prep`, `Modifier.nounMod`, `Modifier.appos`) replacing flat string pairs, the extraction correctly decomposes complex NPs like "the most common element in stars" into `Term.entity(element, [nounMod:element, adj:in])` with the PP object preserved as a `GroundedLexeme`.

6.7 Dev Set Evaluation

We evaluated on 10 unseen examples from the EntailmentBank dev set (2-premise single-step entailments, no slash notation, under 80 characters).

Metric	Count	Rate
All parsed + checked	10/10	100%
Premises with non-opaque Frames	8/20	40%
Hypotheses with non-opaque Frames	3/10	30%
Entailment verified (identity)	1/10	10%
Entailment verified (reasoning)	0/10	0%

Table 4: Dev set evaluation: 10 unseen examples.

The bottleneck is Frame extraction, not reasoning. Parse coverage is 100% but only ∼40% of premises produce non-opaque Frames. The main gaps are:

- Complex transitive VPs ("observes a glowing band")
- Comparative constructions ("less bright than")

- Causative embeddings ("causes X to orbit")
- Conditionals ("if X then Y")
- Gerund NPs ("looking at bright objects")

These correspond to ∼15 additional RGL constructors that need extraction rules. The reasoning rules (inheritance transitivity, substitution, property inheritance, modus ponens) are implemented but have insufficient inputs to fire on these specific examples.

6.8 Structural Entailment Verification

After the Layer 4 redesign, Example 74 verifies via the structural `inheritance_substitution` rule—the correct PLN deduction, not a fallback heuristic:

```
Premise 1: IsA(the sun, star)
Premise 2: Rel(element, hydrogen, star)
Hypothesis: Rel(element, hydrogen, sun)

Rule: inheritance_substitution
  sun <= star (from Premise 1)
  Rel(element, hydrogen, STAR) in Premise 2
  Substitute star -> sun
  Derived: Rel(element, hydrogen, sun)
  Matches hypothesis: VERIFIED
```

This demonstrates the full pipeline operating correctly: English → GF ParseEng → Lean check (115K sig) → RGLView → CopulaOrigin-aware NormClause → Term extraction with typed modifiers → Atom extraction → structural PLN inheritance substitution → **verified**.

The verification uses no heuristics, no entity-bag matching, and no argument permutation. The `inheritance_substitution` rule operates on structured `Term` arguments within `Atom` propositions, performing exact concept matching via `GroundedLexeme.baseName`.

7 Lessons Learned

1. **Use GF's existing ecosystem.** We initially attempted to build custom lexicons and merge `Grammar + DictEng` by hand. The correct entry point is `ParseEng` from `gf-wordnet`, which provides 115,000+ entries with proper morphology and RGL compatibility.
2. **115K functions ≠ 115K semantic rules.** ParseEng is a small structural grammar (∼300 RGL functions) plus a large lexicon (∼115K WordNet senses). Semantic rules cover only the structural part; lexical entries are looked up in a grounding table.
3. **No heuristics.** A suffix-based POS guesser was initially used to build the lexicon, producing ∼30% misclassifications. It was identified in audit

and eliminated. GF's own morphology and WordNet handle vocabulary correctly.

4. **Nested inductives in Lean 4.** The `RawTerm` type with `Array RawTerm` creates a nested inductive that restricts pattern matching. Multi-constructor nested inductives fail; the workaround is a single recursive constructor plus a separate flat type for non-recursive cases.

5. **The C runtime is essential.** The Haskell `gf` CLI is too slow for batch parsing with large grammars (27s/sentence due to per-call PGF loading). The C runtime via Python PGF bindings is 10–100× faster.

8 Concept Grounding and Background Knowledge

8.1 WordNet Synset Grounding

We ground every lexical leaf to its WordNet synset via the `gf-wordnet` concept grounding table—109,184 GF function names mapped to synset IDs, domains, and glosses. Each `ConceptId` carries:

```
structure ConceptId where
  gfFun    : String   -- "star_1_N"
  cat      : String   -- "N"
  synset?  : Option String -- "09467004-n"
  gloss?   : Option String
```

Concept identity prefers synset-level match and falls back to `baseName` when synsets disagree (GF parse may pick the wrong sense without WSD):

$$a \sim b \iff \text{synset}(a) = \text{synset}(b) \ \vee \ \text{baseName}(a) = \text{baseName}(b)$$

8.2 WordNet Hypernymy as Background Knowledge

From the GF WordNet `taxonomy.txt` (119,417 synsets, 89,173 hypernym edges), we extract transitive ancestry chains up to depth 5, yielding **208,907 provable IsA facts** between grounded concepts. Of these, 48,173 are direct parent links; the rest come from transitivity.

Within science domains (chemistry, physics, biology, astronomy, geology, medicine), there are **30,061 provable IsA pairs**—for example:

- `IsA(hydrogen, element)` — *hydrogen is a chemical element*
- `IsA(sun, star)` — *the sun is a star*
- `IsA(whale, mammal)` — *a whale is a mammal*
- `IsA(gasp, breathe)` — *gasping is a form of breathing*

These facts are loaded at runtime as a `Background` structure and injected as additional premises during entailment verification.

8.3 Attr/Rel Unification

Following the council's recommendation (de Paiva), we unified the `Attr` and `Rel` constructors: `Attr(X, P)` is now defined as `Rel(P, [X])`. This eliminates a redundant constructor, removes 16 separate match arms across 4 files, and makes `property_inheritance` a special case of `isa_substitution`.

8.4 Backward-Chaining Proof Search

We replaced the flat rule scan with a fuel-bounded backward-chaining proof search. The verifier decomposes the hypothesis into subgoals and recursively searches premises:

```
partial def prove (premises : Array Atom)
    (fuel : Nat) (goal : Atom)
    : Option (String * Atom) :=
  if fuel == 0 then ... else
  -- 1. Identity
  -- 2. Conjunction introduction (recursive)
  -- 3. IsA transitivity
  -- 4. IsA monotone substitution
  -- 5. Modus ponens (recursive antecedent)
  -- 6. Conjunction elimination
  -- 7. IsA projection
```

Multi-step derivations—e.g., `IsA(A,B) + IsA(B,C) + Rel(P,[C])` deriving `Rel(P,[A])` via two substitution steps—are handled automatically by recursive calls at decreasing fuel.

8.5 Variance Annotations

Each predicate argument position carries a variance annotation: *covariant* (substituting a subtype is sound), *contravariant* (substituting a supertype is sound), or *invariant* (no substitution). For EntailmentBank's generic science facts, the default is covariant. The infrastructure exists for domain-specific overrides—e.g., `Rel(orbits, [earth, sun])` should *not* yield `Rel(orbits, [earth, star])` because argument position 1 is invariant.

9 Herbrand Model Bridge

9.1 From Parsed Text to Model-Theoretic Semantics

The `GFCoreLPBridge` module connects GFCore atoms to the LP kernel's formally verified Herbrand model. The LP kernel (in `Mettapedia.Logic.LP`) provides:

- `T_P_LP`: immediate consequence operator
- `T_P_LP_mono`: monotonicity (kernel-checked)
- `leastHerbrandModel`: via Tarski's fixpoint theorem
- `leastHerbrandModel_db`: EDB facts $\in$ LHM (proven)

All zero sorries—fully kernel-checked.

9.2 The GFCore LP Signature

We define a function-free (Datalog) LP signature:

```
def gfSig : LPSignature where
  constants       := ConceptId  -- 109K concepts
  vars            := String
  relationSymbols := GFRelSym   -- isa + pred(p,n)
  functionSymbols := Empty      -- Datalog!
```

Relation symbols are stratified by arity: `isa` (arity 2) and `pred(p, n)` for each predicate p at arity n. This is standard Datalog practice.

9.3 Translation and Soundness

The `atomToLP` function translates GFCore atoms to LP ground atoms. Higher-order atoms (causes, implies, negation, quantification) return `none`—they stay in the GFCore proof search layer.

Theorem 1 (EDB soundness)**.** *For any list of GFCore atoms and any atom a in that list with $\mathit{atomToLP}(a) = \mathit{some}\ ga$:*

$$ga \in \mathit{leastHerbrandModel}(\mathit{kbFromAtoms}(\mathit{atoms}))$$

This is proven in Lean (zero sorries) as a direct application of `leastHerbrandModel_db`.

9.4 IsA Transitivity as a Single LP Clause

The 48,173 direct WordNet hypernym edges need not be pre-expanded. A single LP clause encodes transitivity:

$$\text{isa}(X, Z) \leftarrow \text{isa}(X, Y), \text{isa}(Y, Z)$$

The T_P operator's fixpoint iteration computes the full transitive closure, yielding all 208,907 provable IsA facts.

10 Native Type Theory via the OSLF Bridge

10.1 From Parse Trees to Modal Type Systems

The authoritative real-GF bridge is now the syntax-only `GFRealSyntaxBridge` module. It connects checked GF trees and generated GF signatures to the `LanguageDef` / OSLF layer without adding hand-authored equations or rewrites. In this lane, GF contributes:

- typed abstract constructors (`GrammarSig`),
- checked trees (`CheckedExpr`),
- multilingual alignment via shared abstract syntax,
- a constructor/category structure on which the OSLF framework can build native-type and modal interfaces.

This is deliberately narrower than treating GF itself as a rewrite calculus. The real pipeline gives us syntax, checking, parsing, linearization, and bilingual abstract-tree agreement; it does not endow GF with a native semantic reduction relation.

The bridge requires only three functions:

```
def gfCheckedExprToPattern : CheckedExpr -> Pattern
def gfFunDeclToGrammarRule : FunDecl -> GrammarRule
def gfSigToLanguageDef : GrammarSig -> LanguageDef
```

`gfCheckedExprToPattern` is the central bridge: checked GF nodes become `Pattern.apply` terms with the GF function name and recursively converted children. The resulting `LanguageDef` contains exactly the categories and constructors exported by the real grammar signature, with zero invented equations and zero invented rewrites.

10.2 Syntax-Only Galois Structure

Even with zero rewrites, the OSLF framework still provides the generic modal adjunction:

Theorem 2 (Galois connection)**.** *For any GF grammar G encoded as a `GrammarSig`:*

$$\Diamond_G \dashv \Box_G$$

where $\Diamond_G(\varphi)(t) \Leftrightarrow \exists q.\, t \rightsquigarrow q \wedge \varphi(q)$ *and* $\Box_G(\varphi)(t) \Leftrightarrow \forall q.\, t \rightsquigarrow q \to \varphi(q)$.

This is proven in Lean (zero sorries) as a direct application of `langGalois`. For the present syntax-only GF lane, the reduction relation is empty, so positive $\Diamond$ witnesses are absent and $\Box$ is vacuous. This is an honest result: the modal structure is available, but it carries no nontrivial reduction content until a separate grounded semantic layer is added above the grammar.

10.3 Native Type Theory

From the Galois connection, the OSLF framework extracts a *native type theory* (NTT): for each GF sort s, a type former classifying terms whose root constructor targets s. This gives every checked GF tree a type-theoretic interpretation:

$$\text{NativeType}(s)(t) \;\Leftrightarrow\; \text{“}t \text{ is a well-formed term of sort } s\text{”}$$

The real content in the current lane is therefore structural rather than reductional: constructor categories, representable sort fibers, and bilingual agreement after checking.

This is the hypercube [11] instantiated at the syntax level for natural language: the current real-GF contribution is the structural face, together with a formally available but vacuous modal adjunction.

10.4 Grounded Real-GF Slices

Two grounded slices are currently in the live lane:

`PaperAmbiguity.` A small generated English/Czech grammar used for exact end-to-end conformance. The authoritative syntax-only `LanguageDef` has 16 categories, 26 constructors, 0 equations, and 0 rewrites. English and Czech witness parses recover the same checked patterns for both telescope readings and both Anna readings.

`project_core.` A larger bilingual abstract grammar exported from the English and Czech RGL family modules. The generated English and Czech JSON exports have identical abstract signatures: 384 abstract functions over 90 categories, with start category `S`. Lowering either export through `gfSyntaxLanguageDef` yields the same syntax-only `LanguageDef`, again with 0 equations and 0 rewrites. On the Rust side, the same artifact is already checked structurally via Lean-exported `language!` fixtures and exact shared-core comparison; compiling the full generated Rust module is presently an engineering-scale issue rather than a semantics issue.

10.5 What NTT Can Honestly Say Today

For the real syntax-only lane, the NTT/OSLF layer presently supports:

- type-theoretic classification by real GF sort;
- bilingual English/Czech agreement after `GFCore.check`;
- constructor-level examples over larger real grammar slices such as `ExistNP`, `QuestCl`/`UseQCl`, and `ProgrVP`;
- explicit proofs and regression checks that the syntax-only lane has no executable reductions.

- exact structural export into the Rust `language!` surface for both `PaperAmbiguity` and the larger `project_core` slice, with canonical shared-core agreement checked mechanically.

What it does *not* yet support is calling those structures "GF rewrites" or presenting $\Diamond$ as semantically inhabited for real GF. Any future nontrivial modal or denotational content must come from a separate grounded layer over real GF artifacts, not by relabeling hand-authored semantic rules as if they were native to GF.

10.6 Typed Datalog for Logic Rules

The `LanguageDef.logic` field now supports typed Datalog clauses via `DatalogClause`, bridged to the LP.Core formalization which provides the T_P immediate consequence operator and the least Herbrand model (via Tarski's fixpoint theorem). This gives `LanguageDef` logic rules strictly stronger semantics than the Rust Ascent compilation: Lean proves the rules are correct, Rust runs them fast.

11 LLM-Guided Gap Filling with Provenance

11.1 Architecture

When the backward-chaining proof search fails, it extracts a *gap diagnostic*: a description of what premise would make the proof succeed, plus the formal atom that is needed.

The gap-fill pipeline:

1. **Proof search** fails with gap: "goal not derivable: `Rel(?verb, <GerundNP>, the eye(pl))`"
2. **LLM oracle** receives the gap, premises, and hypothesis as a structured prompt. Returns a suggested bridging sentence.
3. **GF C runtime** parses the suggestion (via Python PGF bindings, same offline pipeline as premise parsing). Returns `RawTerm` JSON.
4. **Lean consumes** the tree via `FromJson`, `check`, `toRGLView`, `extractSemantics`, `groundAtom`—the identical pipeline used for premises.
5. **Evidence tag**: the new atom is tagged as `EvidenceKind.modelDerived` with source provenance (model name, confidence, prompt).
6. **Proof search retries** with augmented premises.

11.2 Results

On our 10-example dev set, the gap-fill pipeline improves verification from 6/10 to 7/10. Example 2 ("looking at the moon has less of a negative impact on the eyes") is solved:

```
GAP: goal not derivable
LLM (mock, conf=0.5): "the moon is less bright
  than the sun so looking at it has less
  negative impact on the eyes"
GF parsed -> Rel(?verb, the moon, the eye(pl))
  [modelDerived:mock:entailmentbank-gold]
PROVED after LLM fill (identity)
Provenance: GF:ParseEng + mock:entailmentbank-gold
  -> identity
```

The remaining 3 failures (gravity/orbits, weathering/erosion, sound/vacuum) produce well-formed GF parses of the LLM suggestions but the extracted atoms do not match the hypothesis structure. These require domain-specific reasoning rules (causal composition, temporal ordering, negation) beyond the current sequent calculus.

11.3 The Formal System as Ledger

The key architectural insight: **the formal system is the ledger, not the reasoner**. The LLM provides semantic understanding—choosing parses, filling knowledge gaps, suggesting domain rules. The formal pipeline provides:

- **Grammatical certification**: GF ensures the suggestion is syntactically valid English with a well-typed parse tree.
- **Semantic grounding**: every concept links to a WordNet synset; every atom lives in the Herbrand model.
- **Provenance tracking**: `EvidenceKind` tags distinguish empirical facts (WordNet), text interpretations (GF parse), and model-derived suggestions (LLM).
- **Derivation audit**: the proof search records which rule fired and which premises were used.

This enables a trust hierarchy: WordNet taxonomy facts (`empirical`) are trusted; GF-parsed text (`textInterpreted`) is reliable but may have WSD errors; LLM suggestions (`modelDerived`) carry reduced confidence. The PLN BinaryEvidence framework (formalized in mettapedia, 704 theorems) provides the mechanism for propagating confidence through derivation chains.

12 Future Work

Denotational semantic evaluation. A prior formalization effort ("native-grammatical-formalism") built a complete semantic layer over hand-crafted GF encodings: quantified formulas (`QuantifiedFormula2` with $\forall$, $\exists$), variable binding (`StoreAtom` with bind/ref), scope completeness (reachable scope orders = linear extensions), pronoun-binding locality, and evidence-valued denotational semantics (`QEvidenceAtomSem`). This layer has 67

proven world-model theorems and 7 paper-facing semantic highlights, all zero sorries.

The limitation was that it operated on hand-crafted toy grammars, not real ParseEng parses. Now that the pipeline produces real `CheckedExpr` trees from 115K-function ParseEng, the natural next step is to bridge `CheckedExpr` to the OSLF formula layer. The `PGFWitnessIR` module already provides `checkedExprToAbstractNode`, connecting real GF output to the abstract node format used by the semantic layer.

This would give us: English sentence → GF parse → quantified formula → evaluation in an evidence-valued world model—with proven soundness at every step.

Mutual consistency for parse pruning. Given N sentences from a paragraph, parse each through GF and extract grounded atoms. Score the joint interpretation by checking: (a) shared concepts resolve to the same synset, (b) no atom contradicts another in the Herbrand model, (c) how many cross-sentence derivations succeed. Bad parses produce inconsistencies; the consistency score provides a principled parse-selection signal without heuristics.

PLN truth values. Tag atoms with $\langle \text{strength}, \text{confidence} \rangle$ pairs from the formalized PLN evidence framework (BinaryEvidence quantale, 704 theorems). LLM-repaired premises inherit reduced confidence; the proof search propagates evidence through derivation chains.

DBpedia/Wikipedia ingestion. Fetch structured triples from DBpedia's SPARQL endpoint, parse through GF, and add as grounded atoms to the knowledge base. The LP bridge is ready for this.

Auto-ontology formation. The inductive loop: harvest text from authoritative sources, parse via GF, ground to synsets, add to LP knowledge base, identify gaps via failed proof search, fill gaps with LLM-guided repair (at reduced confidence), iterate. This connects to the Hyperseed and WM-PLN programs.

Top-N parsing with PLN disambiguation. Use reasoning success as evidence for parse selection in a PLN factor graph.

Multilingual. GF's architecture guarantees that adding Czech (or any other GF language) requires zero changes to the Lean bridge—only a different concrete syntax at parse/linearize time.

13 Related Work

GF has been used for controlled natural languages in formalization [3], including the SUMO ontology and medical NLP applications. The GF–Haskell and GF–

Java embedded grammars provide typed host-language APIs generated from GF abstract syntax, which inspired our slice generation approach.

EntailmentBank [6] was developed by Allen AI for evaluating multi-hop textual entailment. Existing approaches use neural models for proof generation; our approach is the first (to our knowledge) to use GF parsing with formal verification in Lean.

14 Conclusion

We have demonstrated a complete pipeline from English text to formally verified semantic atoms grounded in a Herbrand model with proven soundness properties—all in Lean 4, with zero sorries in the logic chain.

The pipeline spans five layers: GF parses English (98% coverage on EntailmentBank), GFCore provides the grammar-agnostic trust boundary (25 modules), RGLView/NormClause canonicalize surface variation, the unified Term/Atom layer provides PLN-compatible propositions, and the LP bridge connects to a formally verified least Herbrand model via Tarski's fixpoint theorem.

Concept grounding via 109,184 WordNet synsets and 208,907 provable IsA facts from the WordNet taxonomy provides a substantial ontological foundation. The backward-chaining proof search with variance-annotated substitution gives a principled, extensible inference engine.

The key result: parsed English sentences are *provably members* of the least Herbrand model of the knowledge base—not by heuristic matching, but by kernel-checked theorems composing the T_P operator's fixpoint semantics with the GFCore extraction pipeline.

This lays the foundation for auto-ontology formation: harvesting knowledge from authoritative sources (textbooks, Wikipedia, DBpedia), parsing through GF, grounding to WordNet synsets, and reasoning over the result with formally verified soundness guarantees.

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